

Optimization-driven uncertainty forecasting: Application to day-ahead commitment with renewable energy resources

Sajad Karimi^a, Soongeol Kwon^{b,*}

^a Department of Systems Science and Industrial Engineering, Binghamton University, SUNY, 4400 Vestal Parkway East, Binghamton, NY 13902, USA

^b Department of Industrial Engineering, Yonsei University, 50 Yonsei-ro, Seodaemun-gu, Seoul 03722, Republic of Korea

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ABSTRACT

The participation of power producers who generate electricity from renewable energy resources, e.g., wind and solar, in the electricity market has been significantly promoted by the high penetration of renewable electricity generation. In this case, the unit commitment is required to be formulated and solved to determine the optimal commitment in response to uncertainty existing in renewable electricity generation. To address this problem, a forecasting-first optimization-second approach has been practically applied in that the next-day renewable electricity generation is forecasted first using historical data, and then the unit commitment problem is solved using the forecasted renewable energy generation. However, the forecasting-first optimization-second approach has limitation that forecasting and optimization are decoupled, and thus, forecasting cannot be tuned by reflecting the effect of forecasted renewable energy generation on optimizing unit commitment. Given this context, this study proposes an optimization-driven renewable energy generation forecasting that is designed to integrate the unit commitment problem with the course of regression so that the regression can be done to minimize forecasting error while maximizing profit. Numerical experiments are conducted based on general regression models, e.g., auto-regressive and multiple linear regression models, with various parameter settings, and the results demonstrate that the proposed approach provides better renewable energy forecasting in terms of resulting unit commitment, i.e., greater profit and less penalty, without degrading forecasting accuracy significantly.

1. Introduction

In general, the wholesale electricity market comprises day-ahead and real-time markets that are regulated by the government and Independent System Operators (ISOs). Day-ahead wholesale electricity markets have been set up to secure electricity supply while reducing volatility [1], and participants in this market must announce their hourly electricity supply commitment for the next day. At the same time, the real-time market is supposed to compensate mismatch between actual demand and committed supply. Due to the high penetration of renewable electricity generation, the participation of power producers who generate electricity from renewable energy resources, e.g., wind and solar, in the electricity market has significantly increased [2]. Specifically, renewable electricity generation is encouraged to be scheduled and committed in the day-ahead market by the various rules implemented in electricity markets [3]. Because day-ahead prices are higher than real-time prices on average while there will be a penalty charged on shortage in commitment [4], day-ahead commitment is required to be determined to maximize revenue in

response to uncertain renewable electricity generation. Given this context, this study considers the case where power producers generate electricity solely from renewable energy resources and sell it in the electricity market through day-ahead commitment. Note that because power producers relying on renewable electricity generation tend to be a price taker in that they are willing to sell generated electricity at the price cleared by the market, rather than bidding a price [5], this study considers an optimization problem formulated to find an optimal day-ahead commitment that maximizes profits given day-ahead prices while considering real-time prices as well as penalty cost that will be charged if the commitment cannot be met. Although there has been a body of literature that proposes stochastic optimization models that consider specific distribution or sampled data to incorporate stochastic renewable electricity generation with optimizing day-ahead unit commitment [6–8], renewable electricity generation forecasting would be expected to play a significant role to optimize day-ahead commitment because optimization over forecasted renewable electricity generation

* Corresponding author.

E-mail addresses: skarimi1@binghamton.edu (S. Karimi), sg.kwon@yonsei.ac.kr (S. Kwon).

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has still been considered in real-world practice [9], which can get generalized as “forecasting-first optimization-second” approaches.

Given this, this study is particularly motivated to improve the current practice of renewable electricity generation forecasting toward its application in the day-ahead commitment problem based on the forecasting-first optimization-second scheme. The main idea is to reflect the feedback from the optimization on the course of regression so that the coefficients of the regression model can be fitted to minimize forecasting error and optimality gap simultaneously. Note that the optimality gap denotes the difference between the objective function value found by the optimization model given forecasted solar electricity generation and the objective function value found by the optimization model given actual solar electricity generation where the objective function value denotes the total profit derived by the day-ahead commitment and real-time trading. Because forecasted renewable electricity generation would ultimately be used as an input parameter for optimization model based on forecasting-first optimization-second approaches, solution space of day-ahead commitment problem would be specified by forecasted solar electricity generation, which means the quality of resulting solution would be depending on forecasting. Hence, this study intends to develop optimization-driven forecasting where regression can be done by reflecting the feedback of optimization model. To validate this idea, this study considers general regression models that can be used to forecast next-day solar electricity generation and modifies loss function used in regression so that the regression models’ coefficients can be fitted to minimize forecasting error and optimality gap simultaneously. It is reasonably expected to derive better day-ahead commitment by adopting our proposed approach compared with the conventional forecasting-first optimization-second approach where forecasting and optimization are decoupled and conducted separately without any feedback. It should be emphasized that the regression model in decoupled approach cannot be easily tuned in a similar way because the impact of forecasting error on day-ahead commitment cannot be directly captured and reflected in regression phase because day-ahead commitment is required to be optimized in response to day-ahead and real-time price. To investigate whether the proposed approach can be practically applied, this study mainly focuses on showing how existing regression modes can be explicitly modified by applying the proposed approach and design numerical experiments to solely evaluate the performance improvement achieved by the proposed approach when it is applied to both regression models.

The following sections comprise the remainder of the paper: Section 2 summarizes relevant studies and emphasizes the paper’s uniqueness and contributions. Section 3 discusses the day-ahead electricity market in broad terms and then delves into the details of our proposed model. In Section 4, mathematical optimization models and forecasting techniques are presented. Section 5 examines numerical experiments to verify our proposed model, while Section 6 concludes the study and discusses future work.

2. Literature review

Forecasting the day-ahead renewable energy generation is the main approach that can be used to generate a deterministic energy generation parameter for the mathematical optimization model of the unit commitment problem. In an ideal situation, forecasting with 100% accuracy leads to an optimal solution. However, in the real world, there is always a forecasting error that causes less profit in the energy commitment problem. There are several forecasting techniques for renewable energy generation at various resolutions, e.g., hourly, daily, etc.; planning horizons, e.g., day-ahead, hour-ahead, etc., and input data. In terms of the source of input data, the literature identifies two distinct approaches: endogenous forecasting, which relies exclusively on time series data from renewable energy generation, and exogenous forecasting, which incorporates external data such as Numerical Weather Prediction (NWP) information, such as temperature, humidity, radiation, wind speed, etc., satellite images, and local measurements. Statistical and machine learning models such as Auto-Regressive (AR), Auto-Regressive Moving Average (ARMA), Auto-Regressive Integrated Moving Average (ARIMA), regression, Neural Networks (NN), Support Vector Machine (SVM), etc. are popular models to predict renewable energy generation. Because solar and wind power are the two major renewable energy sources [19], we review the existing prediction models for those sources. Table 1 shows the list of several studies that propose a forecasting method for renewable energy generation. In this regard, [10] presented an Exogenous Input based Statistical model (ARX-ST) to forecast the solar power generation. In their model, spatial and temporal correlations among neighboring sites were considered. They compared the performance of their model with conventional persistence (PSS), back-propagation neural network (BPNN), and AR models. They demonstrated that their proposed model outperformed existing models in 1-h and 2-h ahead forecasting. [11] proposed a method for day-ahead solar energy forecasting based on the daily weather forecast. First, three single-stage models, i.e., Multiple Linear Regression (MLR), Generalized Regression Neural Network (GRNN), and Feed Forward Neural Network (FFNN), were designed. Then, a stepwise regression method was applied to select the best meteorological variables. By combining the variable selection and single-stage models, three hybrid models were developed. Accuracy metrics indicated that hybrid models provide better forecasting results. [12] performed an Artificial Neural Network (ANN) to predict hourly day-ahead solar generation based on the weather prediction data. Also, they conducted a sensitivity analysis for the accuracy of their model based on the size of the training dataset. [16] proposed three models based on ARMA, ANN, and Adaptive Neuro-Fuzzy Inference System (ANFIS) to predict wind power for different short term forecasting horizons. The simulation result showed that ANFIS and ARMA are suitable for 10-minute and 1-hour ahead forecasting, respectively. [17] presented an improved Long Short-Term Memory (LSTM) neural network for predicting wind power. Their model improved the effect of forget-gate and modified the activation function to speed up the convergence. They showed that their method could increase the accuracy by 18.3% compared to other existing methods. [18] used two time series prediction

Table 1
Table of references of renewable energy generation forecasting.

Reference	Renewable energy	Method	Forecasting horizon	Resolution
Yang et al. [10]	Solar	ARX-ST, PSS, BPNN, AR	5 min, 15 min, 1 h, 2 h	1 min, 1 h
Ramsami et al. [11]	Solar	MLR, GRNN, FFNN	24 h	1 h
Leva et al. [12]	Solar	ANN	24 h	1 h
Kardakos et al. [13]	Solar	SARIMA, ANN	day-ahead	1 h
Shi et al. [14]	Solar	SVM	day-ahead	15 min
Gigoni et al. [15]	Solar	KNN, ANN, SVR, QRF	day-ahead	1 h
Chen and Folly [16]	Wind	ARMA, ANN, ANFIS	10 min, 1 h	10 min
Yu et al. [17]	Wind	LSTM	6 h	30 min
Cadenas et al. [18]	Wind	ARIMA, NARX	10 min, 1 h	10 min, 1 h

models, univariate ARIMA and Nonlinear Auto-Regressive Exogenous Artificial Neural Network (NARX), to forecast the wind power and showed that NARX performs better than ARIMA for one-step-ahead forecasting.

Forecasting-first optimization-second is an approach in which the renewable energy generation is forecasted at the first step. Then the forecasted value of day-ahead energy generation is used in an optimization model to determine the optimal energy commitment. [4] proposed a forecasting-first optimization-second approach for wind trading in the day-ahead market. Kernel density estimation was used to create a probabilistic wind power forecasting model. Uncertainty in day-ahead and real-time prices are assumed to be Gaussian. After forecasting the wind power, the optimal day-ahead commitment was obtained for different risk preferences and penalty costs. The model was tested based on the real dataset of United States wind farms. [5] addressed the “solar commitment problem” to determine the solar energy that will be sold in the next day in the electricity market to maximize the revenue. They used forecasted solar generation and weather measurement to create a conditional probability distribution of next-day generation. To forecast solar energy, they used a “black box” model, called Solar-TK, that is an open-source tool that predicts the amount of solar power generated based on the location and meteorological information.

According to the reviewed studies in this section, the traditional forecasting-first optimization-second approach neglects the optimization knowledge, i.e., the objective function, electricity prices, penalty cost, etc., in the first step. In this paper, we propose an optimization-driven forecasting model to predict the renewable energy generation for the day-ahead unit commitment problem by leveraging a mathematical optimization model. To be more specific, we integrate a forecasting loss function and a mathematical optimization model throughout the training process to create a tailored prediction model for the energy generation in the unit commitment problem. The proposed model attempts to minimize forecasting error and maximize profit from day-ahead energy trading during the training process. Therefore, it is expected that the obtained forecasting coefficients will lead to a better optimal solution in the test dataset.

3. Problem description

In terms of the paper’s primary objective, which is to propose a novel forecasting approach for the day-ahead unit commitment problem, we first explain the problem in general and then develop our proposed model by comparing it to current forecasting methods.

3.1. Day-ahead commitment with renewable energy resources

Power Producers as participants in the day-ahead electricity market are required to determine their hourly energy supply commitments for the next day, and the day-ahead price is determined based on the bids submitted by power producers using a uniform price multiunit auction [21]. The day-ahead energy generation for the traditional fuel-based generators is very accurate. The traditional producers base their bid price on the current fuel price and well-known trends, such as holidays, weekends, and weekdays. In comparison, because renewable energy providers incur no fuel costs and few other operational expenses, their best offer is at or near level 0. Thus, renewable energy generators are price takers, prepared to sell energy at almost any positive price [22]. On the other hand, operators can sell or buy energy during the day in real-time electricity markets. The real-time energy market is created to balance the differences between day-ahead energy commitment and actual demand. In this paper, we assume that the day-ahead prices are known, and commitments are announced at the end of the previous day. Also, curtailment is not considered. It means that the output of renewable generation is not deliberately deducted. Operators may generate more or less than their commitment. If there is a shortage in supply, the operator incurs a penalty cost that is a percentage of the day-ahead price. In contrast, if the operator generates more energy than the committed amount, it can sell it in the market based on the “real-time price”. In a deterministic situation, because solar energy generation has no cost, the optimal value of energy commitment for solar operators depends on day-ahead and real-time price. If the day-ahead price is greater than the real-time price, energy commitment should be as much as generated solar energy. Otherwise, the operator should commit zero energy supply and sell the generated energy in the real-time market. In real practice, there is uncertainty about the energy supply by solar operators because the solar energy generation for the next day is not deterministic. Furthermore, the real-time price of the next day is not determined in the current day. Therefore, the exact optimal amount of commitment for the next day cannot be computed. In that case, the day-ahead predicted value of solar energy generation is used to determine the energy supply commitment. Typically, the day-ahead price is greater than the real-time price in the energy market. Fig. 1 shows a box plot of average hourly day-ahead and real-time price for the Midcontinent Independent System Operator (MISO) [20] for four months. The average day-ahead price is greater than the real-time price at all hours of the day. As that situation is very common, generally power producers try to maximize their profits by committing renewable energy generation as much as possible through the day-ahead market rather than the real-time market. In that situation, adopting accurate forecasting methods to predict renewable energy generation would be

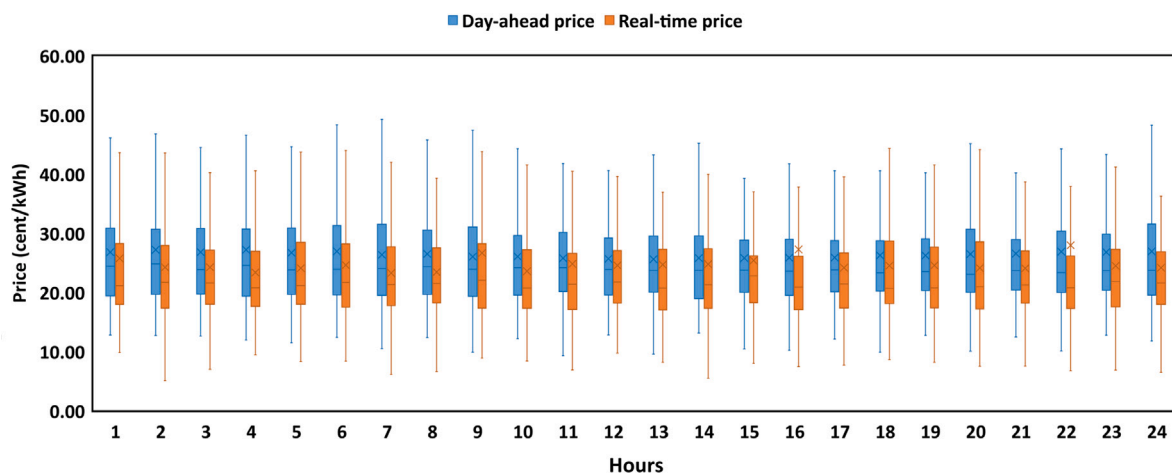
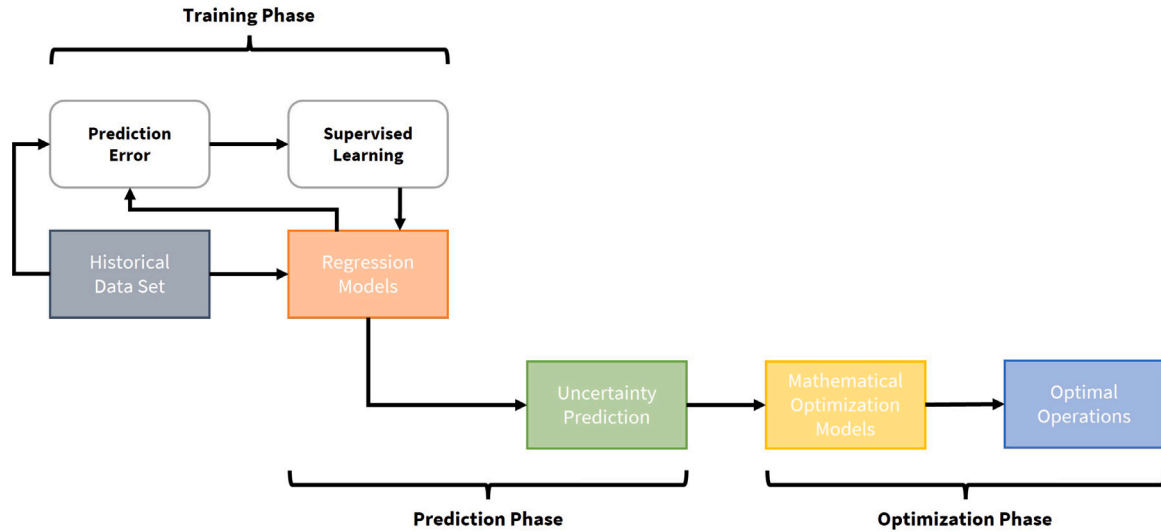
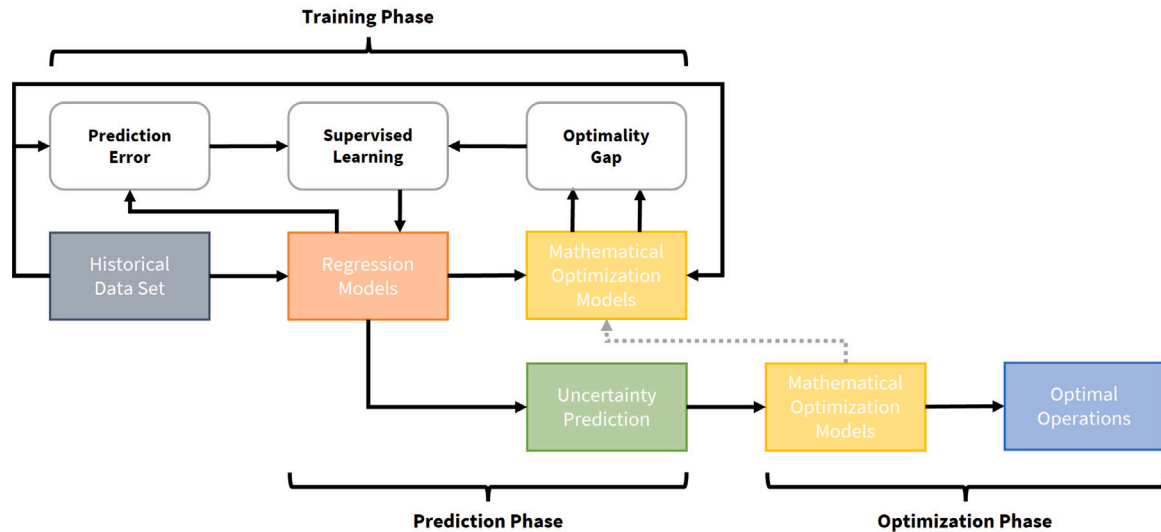


Fig. 1. Hourly day-ahead and real-time prices in wholesale energy market [20].



(a) A schematic diagram for forecasting-first optimization-second approach



(b) A schematic diagram for optimization-driven forecasting

Fig. 2. Schematic diagrams of traditional and our proposed forecasting approaches.

crucial. As a result, we are interested in the day-ahead unit commitment associated with renewable energy generation, with a particular emphasis on the development of more accurate forecasting methods.

3.2. Optimization-driven forecasting

In the unit commitment problem with renewable energy generation, historical data can be used to predict the value of renewable energy. A typical approach that has been used for these types of problems is forecasting-first optimization-second, which is indicated in Fig. 2(a). Based on that approach, a regression method is used to create a forecasting model based on historical data of renewable energy in the training phase. That approach's training phase is completely separate from the optimization phase and is only concerned with minimizing prediction error. After obtaining the forecasting model, i.e., the model's coefficients, the hourly renewable energy generation for the next day is predicted. Then, as a deterministic parameter, day-ahead predicted renewable energy is imported into the optimization model, and the optimum day-ahead energy is determined. The primary disadvantage of that strategy is that the optimization model's information is not

employed during the training phase. The main idea behind the proposed approach is to reflect the optimality gap over the course of regression as shown in Fig. 2(b) so that the regression models' coefficients can be fitted to minimize both forecasting error and optimality gap simultaneously. It is reasonably expected to derive better day-ahead commitment by adopting our proposed approach compared with the conventional forecasting-first optimization-second approach where forecasting and optimization are decoupled and conducted separately without any feedback. This is because solution space searched during the course of finding an optimal solution will be specified by forecasted data, and thus, forecasting should be tuned by reflecting this aspect to find a better solution. In other words, the impact of forecasting on optimizing day-ahead commitment cannot be directly reflected on the regression when forecasting and optimization are decoupled. To address this limitation, this study intends to formulate the course of regression to find the optimal value of coefficients that returns both the forecasting errors as well as the optimality gap by reflecting the feedback from the optimization model to the regression as described in Fig. 2(b), and this would be the main contribution of our study.

4. Mathematical models

This section presents the mathematical optimization models for day-ahead commitment problems, conventional forecasting models, and our proposed optimization-driven forecasting models.

4.1. Notation

• Sets and Indices

- t : Index set of discrete one-hour time periods $t \in T = \{1, 2, \dots\}$.
- h : Index set of discrete one-hour time periods for one day $h \in H = \{1, \dots, 24\}$.
- U_h : Set of time periods t that are in h th hour of the day.
- l : Index set of AR coefficients $l \in L = \{1, 2, \dots, N^L\}$ where N^L is the number of AR coefficients.
- k : Index set of independent variables $k \in K = \{1, 2, \dots, N^K\}$ where N^K is the number of independent variables.
- i : Index set of train samples $i \in I = \{1, 2, \dots, N^I\}$ where N^I is the number train samples (days).
- j : Index set of test samples $j \in J = \{1, 2, \dots, N^J\}$ where N^J is the number test samples (days).

• Parameters

- DP_t : Day-ahead price at time $t \in T$.
- RP_t : Stochastic parameter for real-time price at time $t \in T$.
- PC_t : Penalty cost for shortage to fulfill the day-ahead commitment at time $t \in T$.
- S_t : Stochastic parameter for renewable energy generation at time $t \in T$.
- $\hat{S}_t$: Predicted value of renewable energy generation at time $t \in T$.
- $V_{k,t}$: The value of k th independent variable at time t .

• Decision Variables

- x_t : Day-ahead renewable energy commitment at time $t \in T$, where $x_t \geq 0$.
- y_t : Mismatch between actual renewable energy generation and day-ahead commitment at time $t \in T$, where y_t is unrestricted in terms of sign. For example,
 - * $y_t > 0$ if actual renewable energy generation is more than day-ahead commitment.
 - * $y_t < 0$ if day-ahead commitment is more than actual renewable energy generation.
 - * $y_t = 0$ if day-ahead commitment is the same as actual renewable energy generation.
- y'_t, y''_t : Surplus and shortage of renewable energy generation to meet day-ahead commitment at time $t \in T$, respectively.
- α^h and β_l^h : AR coefficients for the h th hour of the day.
- α and β_k, t : MLR coefficients.

4.2. Optimization model for day-ahead unit commitment problem

The problem can be formulated as follows to determine optimal day-ahead commitment, x_t , while considering penalty cost PC_t , real-time price RP_t and renewable energy generation S_t :

$$\max \sum_{t \in T} DP_t x_t + RP_t y'_t - PC_t y''_t \quad (1a)$$

$$\text{s.t. } y_t = S_t - x_t \quad \forall t \in T \quad (1b)$$

$$y_t = y'_t - y''_t \quad \forall t \in T \quad (1c)$$

$$y' \leq S_t \quad \forall t \in T \quad (1d)$$

$$x_t \geq 0 \quad \forall t \in T \quad (1e)$$

$$y_t \text{ unrestricted } \forall t \in T \quad (1f)$$

$$y'_t, y''_t \geq 0 \quad \forall t \in T \quad (1g)$$

The objective function (1a) comprises the profit from day-ahead commitment, the expected profit from real-time trading, and the penalty for the shortage to fulfill the day-ahead commitment given real-time price, renewable energy generation, and penalty cost. Constraints (1b) and (1c) determine the surplus, y'_t , and shortage, y''_t , of renewable energy generation to fulfill the day-ahead commitment, x_t , in response to the realized renewable energy generation, S_t . Constraints (1d)–(1g) are the sign constraints for the decision variables.

In real-world practice, day-ahead price and penalty cost are known and renewable energy generation and real-time price are unknown when the problem is solved. According to the forecasting-first optimization-second approach, the value of renewable energy generation can be forecasted, at the first step, and then imported to this optimization model, at the second step, in order to obtain the value of x_t .

4.3. Forecasting models

In this study, we use AR and MLR as forecasting models for the endogenous and exogenous approaches, respectively. Those models can be used in the first step of the forecasting-first optimization-second technique or as the basis for our proposed optimization-driven forecasting approach. AR, as an endogenous approach, is a time series forecasting method that uses historical renewable energy generation data without external data. Conversely, MLR, as an exogenous approach, uses external data, e.g., weather forecasting and historical data of renewable generation, to forecast next-day renewable energy generation.

4.3.1. Auto-regressive model

As an endogenous approach, AR is used to forecast renewable energy. AR is a well-known algorithm used as a time-series forecasting method. According to that method, renewable energy generation at each hour can be predicted based on a linear relationship between N^L previous hours:

$$\hat{S}_t = \alpha + \sum_{l \in L} \beta_l S_{t-l}, \quad (2)$$

where $\hat{S}_t$ is the predicted renewable energy generation at time period t , and α and β_l are the coefficients of the AR method, respectively. In that case, the renewable energy generation of each hour is predicted based on its previous N^L hours. Eq. (2) is used to predict one-hour ahead renewable generation while in the unit commitment problem, we need to predict all the next day's hours at the beginning of the day. Therefore, we need to perform multi-step ahead time-series prediction which is more difficult than a single-step prediction, because in this kind of forecasting the errors are accumulated and the accuracy is reduced [23]. There are two main approaches to predict multi-step ahead in time series forecasting: iterative and direct multi-step ahead forecasting [24]. In the iterative approach, we forecast the renewable energy of the first hour to the 24th hour sequentially in 24 steps. The predicted renewable energy in each step can be used as an input for the next step. However, in the direct approach, 24 forecasting models with different coefficients are created to predict renewable energy generation of 24 h of the next day. In this study, we use the direct approach. Therefore, Eq. (2) can be written as:

$$\hat{S}_t = \alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l} \quad \forall h, t \in U_h, \quad (3)$$

α^h and β_l^h are AR coefficients for the h th hour of the day. For the training process, actual renewable energy generation of N^I days ($N^I \times 24$ hours) and their previous N^L hours is used to find AR coefficients

by solving a mathematical optimization model to minimize the loss function. Sum of squared errors (4) and the sum of absolute errors (5) are the two most used loss functions in AR:

$$\min Z^h = \sum_{t \in U_h} (S_t - (\alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l}))^2 \quad \forall h, \quad (4)$$

$$\min Z^h = \sum_{t \in U_h} |S_t - (\alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l})| \quad \forall h. \quad (5)$$

In this study, we use the sum of absolute errors as the loss function of the AR. After obtaining the AR coefficients (α^h and β_l^h) in the training process, the 24-hour ahead renewable energy generation for every given day can be forecasted. The forecasting process is applied to predict the renewable energy of N^J days ($N^J \times 24$ hours) as a test dataset.

4.3.2. Multiple linear regression model

For exogenous forecasting, we consider MLR to find a linear relationship between multiple independent variables such as day-ahead weather predictions and response variables (renewable generation) in each hour of the next day. Based on this model, the predicted renewable generation can be obtained by the following equation:

$$\hat{S}_t = \alpha + \sum_{k \in K} \beta_{k,t} V_{k,t}, \quad (6)$$

where $V_{k,t}$ is the value of the k th independent variable at time t , α is the constant coefficient of MLR, and $\beta_{k,t}$ is the coefficient of variable k at time t . Note that, in this model, the same set of coefficients is used for every hour of the day. MLR model tries to find the optimal value of coefficients of Eq. (6) (α and $\beta_{k,t}$) by minimizing the loss function in the training process. Eqs. (7) and (8) show the squared and absolute loss functions for MLR, respectively:

$$\min Z = \sum_{t \in I} \sum_{k \in K} (S_{k,t} - (\alpha + \sum_{k \in K} \beta_{k,t} V_{k,t}))^2. \quad (7)$$

$$\min Z = \sum_{t \in I} \sum_{k \in K} |S_{k,t} - (\alpha + \sum_{k \in K} \beta_{k,t} V_{k,t})|. \quad (8)$$

4.3.3. Traditional forecasting-first optimization-second

The main goal of the regression models is to find the best values of regression coefficients in order to minimize the forecasting error, which is the difference between actual and predicted renewable energy generation. In the traditional forecasting-first optimization-second approach, AR and MLR are used to find the value of next-day renewable energy generation at the first step. This approach only focuses on minimizing the forecasting error without considering the profit of day-ahead unit commitment. We consider the absolute error (Eqs. (5) and (8)) as loss function in this approach because they can be linearized, and the coefficients can be obtained by solving a linear mathematical optimization model. After predicting the next-day renewable energy generation by AR or MLR, the predicted value is imported to the Model (1) as S_t parameter and the value of x_t is obtained by solving this model.

4.4. Optimization-driven forecasting

Unlike the regression models, in our proposed approach we incorporate the mathematical optimization model of day-ahead unit commitment in the forecasting models. Using our proposed models, forecasting error and optimality gap are simultaneously minimized while the regression models minimize only the forecasting error. Two models are proposed based on endogenous and exogenous approaches by combining the forecasting method and the day-ahead unit commitment optimization model. For the endogenous approach, we use AR as a basis and construct our proposed model by combining its absolute loss function (Eq. (5)) and the mathematical model of the unit commitment

problem (Model (1)). Model (9) shows our proposed optimization-driven forecasting model for the endogenous approach. This model is solved based on the train datasets of day-ahead price, real-time price and renewable energy generation. The AR coefficients (α and β) are obtained by solving this model such that the optimality gap and forecasting error are minimized simultaneously for the training dataset. Based on the obtained AR coefficients, we can forecast the renewable energy generation for any given day by using Eq. (3) and set the committed day-ahead equal to the predicted renewable energy generation. Similar to the AR, a directing multi-step approach is used in our proposed model. Therefore, the model is applied 24 times to extract prediction coefficients for each of the 24 h.

$$\min Z^h = \sum_{t \in U_h} \left(W_1 \left((DP_t S_t) - (DP_t (\alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l}) + RP_t y'_t - PC_t y''_t) \right) + W_2 \left| S_t - (\alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l}) \right| \right) \quad (9a)$$

$$\text{s.t. } y_t = S_t - (\alpha^h + \sum_{l \in L} \beta_l^h S_{t-h-l}) \quad \forall t \in U_h \quad (9b)$$

$$y_t = y'_t - y''_t \quad \forall t \in U_h \quad (9c)$$

$$y'_t \leq S_t \quad \forall t \in U_h \quad (9d)$$

$$\alpha^h \text{ unrestricted} \quad (9e)$$

$$\beta_l^h \text{ unrestricted} \quad \forall l \in L \quad (9f)$$

$$y'_t, y''_t \geq 0 \quad \forall t \in U_h \quad (9g)$$

Objective function (9a) minimizes the optimality gap and forecasting error. W_1 and W_2 are regularization parameters that may be used to adjust the trade-off between the optimality gap and forecasting error. By increasing the ratio of W_1/W_2 , the model increases the effect of the optimality gap on the result compared to the forecasting error. If we set $W_1 = 0$ and $W_2 > 0$, the proposed model turns into the conventional AR. Constraints (9b) and (9c) determine the surplus, y'_t , and shortage, y''_t , of renewable energy generation to fulfill day-ahead commitment. Constraint (9d) ensures that the surplus of the renewable generation is less than an actual renewable generation.

Exogenous models assume that the next-day hourly weather forecast is available when the power producer needs to commit to supplying a specified amount of energy. Our proposed model for exogenous forecasting combines the MLR forecasting method and unit commitment optimization model to obtain better results in terms of profit. The MLR loss function (Eq. (8)) and Model (1) are integrated to construct our proposed optimization-driven forecasting model for exogenous approach as follows:

$$\min Z = \sum_{t \in T} \left(W_1 \left((DP_t S_t) - (DP_t (\alpha + \sum_{k \in K} \beta_{k,t} V_{k,t}) + RP_t y'_t - PC_t y''_t) \right) + W_2 \left| S_t - (\alpha + \sum_{k \in K} \beta_{k,t} V_{k,t}) \right| \right) \quad (10a)$$

$$\text{s.t. } y_t = S_t - (\alpha + \sum_{k \in K} \beta_{k,t} V_{k,t}) \quad \forall t \in T \quad (10b)$$

$$y_t = y'_t - y''_t \quad \forall t \in T \quad (10c)$$

$$y'_t \leq S_t \quad \forall t \in T \quad (10d)$$

$$\alpha \text{ unrestricted} \quad (10e)$$

$$\beta_{t,k} \text{ unrestricted} \quad \forall t \in T, k \in K \quad (10f)$$

$$y'_t, y''_t \geq 0 \quad \forall t \in T \quad (10g)$$

Objective function (10a) contains two parts to minimize the optimality gap and forecasting error simultaneously regarding the regularization parameters, W_1 and W_2 . Constraints (10b) and (10c) computes the surplus, y'_t , and shortage, y''_t , of renewable energy generation regarding

the day-ahead energy commitment. Constraint (10d) guarantees that the surplus of the renewable energy generation at time t is always less than or equal to the actual renewable energy generation at time t .

Our proposed model can be trained with the training dataset in order to find the best value of α and $\beta_{k,t}$. The renewable energy generation of the next day can be computed based on the obtained value of α and $\beta_{k,t}$, and the value of independent variables, e.g., predicted weather information, for each hour of the next day by using Eq. (8).

5. Numerical experiment

This section evaluates our proposed models by conducting several numerical experiments to compare their forecasting error and optimality gap with the conventional endogenous and exogenous forecasting models. We use historical data of solar power generation, day-ahead, and real-time electricity price and apply the proposed models and conventional methods on the data to determine the day-ahead solar energy commitment in various settings. The sizes of training and testing data used in numerical experiments are 300 days and 100 days, respectively, and the proposed optimization model is formulated to fit regression models' coefficients that will be used to forecast daily solar electricity generation day-ahead. Although there exists seasonality in data due to the changes in weather conditions according to the changes in seasons, it would be gradual and not significant in daily forecasting because seasonality showed to be lasting over a longer time period. Hence, both regression models have not been modified specifically to reflect seasonality considering the scope of the problem and the size of data in this study.

5.1. Experiment design

Various combinations of regularization parameters (W_1 , W_2) and penalty cost rates are investigated in the numerical experiments to determine their effect on forecasting error and optimality gap. Penalty cost rate varies in different markets. [4] considers 0%–100% of the day-ahead price as penalty cost rate respectively. This numerical experiment analyzes the results based on 11 different levels for the penalty cost rate between 0% to 100% of the day-ahead price. To increase the model's accuracy, night hours with no sun radiation are excluded from datasets. In this case, each day contains 12 h (from 9:00 am to 9:00 pm) with solar radiation. For measuring the forecasting error, Normalized Root Mean Square Error (nRMSE) is used:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{S}_i - S_i)^2}, \quad (11)$$

$$\text{nRMSE} = 100 \times \frac{\text{RMSE}}{\bar{S}}, \quad (12)$$

where $\bar{S}$ is the average actual solar generation value for the test dataset. Furthermore, the optimality gap is calculated based on the following equation:

$$G = \sum_{i \in T} (\max\{DP_i, RP_i\} S_i) - (DP_i \hat{S}_i + RP_i \max\{0, S_i - \hat{S}_i\} - PC_i \max\{0, S_i - \hat{S}_i\}). \quad (13)$$

Day-ahead and real-time electricity prices are obtained from the Mid-continent Independent System Operator (MISO) [20]. We consider solar energy as the renewable energy source in this numerical experiment. Solar power data is obtained from Global Energy Forecasting Competition 2014 (GEFCom2014), which is collected from solar plants in Australia [25].

The solar data contains hourly solar power generation and hourly day-ahead forecasted meteorological information collected from European Centre for Medium-Range Weather Forecasts (ECMWF). Therefore, this dataset can be used for endogenous and exogenous forecasting. Table 2 shows the response variables (solar power) and 12

Table 2

Independent variables for exogenous forecasting.

Variable name	Units
Total column liquid water (tclw)	kg m ⁻²
Total column ice water (tcIW)	kg m ⁻³
Surface pressure (SP)	Pa
Relative humidity at 1000 mbar (r)	%
Total cloud cover (TCC)	0–1
10-m U wind component (10u)	m s ⁻²
10-m V wind component (10v)	m s ⁻²
2-m temperature (2T)	K
Surface solar radiation downward (SSRD)	J m ⁻²
Surface thermal radiation downward (STRD)	J m ⁻³
Top net solar radiation (TSR)	J m ⁻⁴
Total precipitation (TP)	m

independent variables of the dataset. The value of solar power generation in this dataset is a number between 0 to 1, which shows the ratio of the generated solar power to the maximum capacity of solar panels. In this numerical experiment, we set the maximum capacity of the panels as 30 MW. We divide the solar power, day-ahead, and real-time price into two sets, three months as the training and one month as the testing datasets.

5.2. Numerical analysis based on endogenous forecasting

For endogenous forecasting, the conventional AR model and our proposed endogenous forecasting model are applied to the training and testing data. AR is trained with a dataset of the solar power generation (without weather information) to obtain AR coefficients, while our proposed model needs electricity price data besides solar power generation for the training process. As we use the multiple-step direct method to forecast the next 12 h, AR and our proposed model are trained 12 times. Therefore, 12 sets of coefficients are obtained to predict solar power generation of each hour in the next day. The most important assumption for building an AR model is that the time series data is stationary, which means the statistical properties of the time series do not change over time. We use the Augmented Dickey–Fuller (ADF) test to check the stationarity of the solar power data. In the ADF test, the null hypothesis is “the data is stationary”, and the alternative hypothesis is “the data is non-stationary”. After applying the ADF test on the solar power data, the p -value is 0.0045. Because the p -value is less than 0.05, we reject the null hypothesis. Therefore, the solar power data is stationary, and the AR model can be applied to this data.

In the first analysis, we set the penalty cost rate at 50% as a sample to compare AR and our proposed model in forecasting error and optimality gap. Table 3 shows the value of nRMSE and the optimality gap of AR and our proposed model. To assess the effect of regularization parameters on forecasting error and optimality gap, the proposed model is solved using different combinations of the W_1 and W_2 . When we apply AR to forecast the day-ahead solar power generation, we obtain a better result than our proposed model in terms of the forecasting error. However, the optimality gap of AR is larger than our model. The result sounds reasonable because the main focus of AR is reducing the forecasting error without considering the optimization aspect, while our proposed model considers both forecasting and optimization aspects. The result shows that our model (with $W_1 = 1$ and $W_2 = 20$) can provide an almost similar nRMSE but a better optimality gap. This is an important result because it indicates a possibility of increasing electricity commitment profit without significantly changing the forecasting error. Fig. 3 shows the chart of nRMSE and optimality gap for AR and our proposed model in various combinations of W_1 and W_2 when the penalty cost rate is 50%. In the proposed model, by increasing the ratio of W_1/W_2 , the rRMSE increases, and the optimality gap decreases.

Fig. 4 compares the actual and predicted solar power generation by AR and our proposed model for 7 sample days in the test dataset

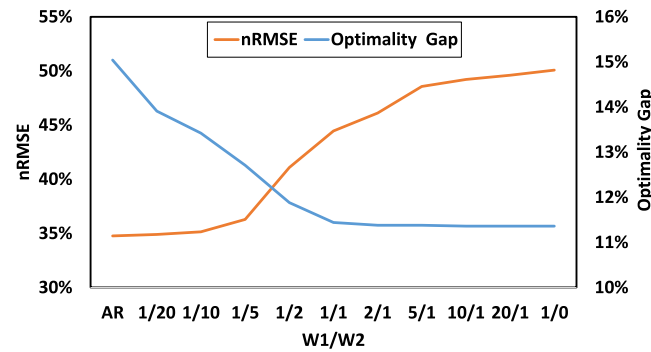


Fig. 3. The nRMSE and optimality gap of AR and proposed model in various ratios of W_1/W_2 when the penalty rate is 50%.

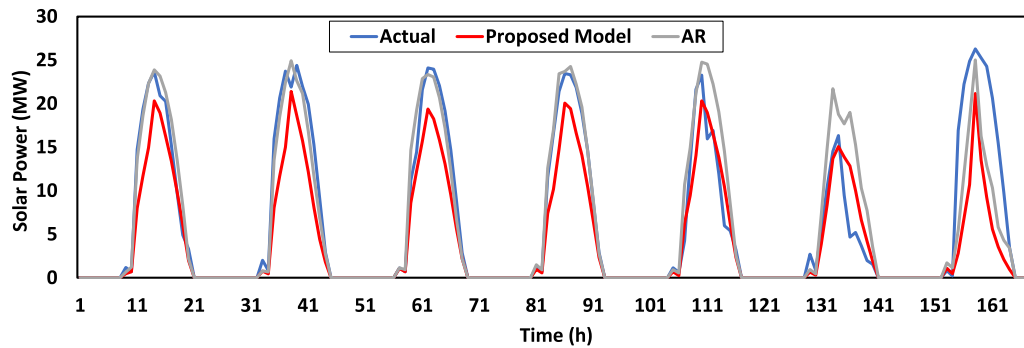
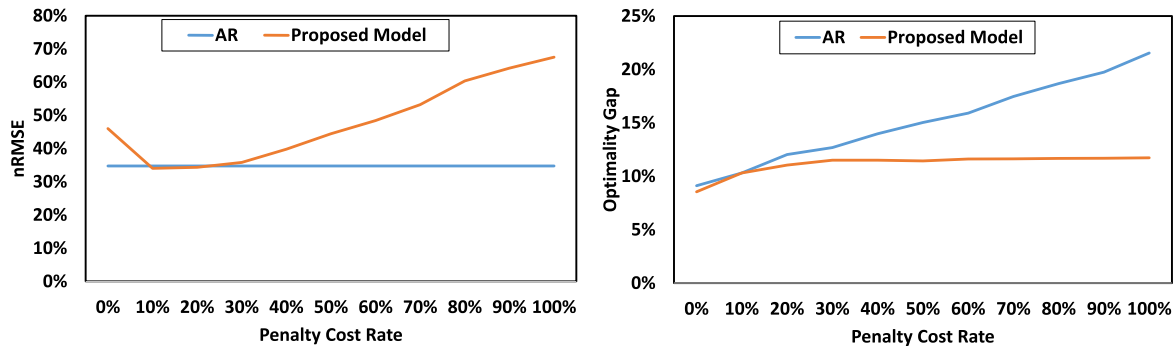


Fig. 4. Actual and predicted solar power generation by AR and our proposed model when $W_1 = W_2 = 1$ and penalty rate is 50%.



(a) nRMSE of AR and proposed model in various penalty rate when $W_1 = 1$ and $W_2 = 1$

(b) Optimality gap of AR and proposed model in various penalty rate when $W_1 = 1$ and $W_2 = 1$

Fig. 5. Comparison of forecasting error and optimality gap of AR and our proposed model.

Table 3
Comparison between AR and our proposed model when penalty cost rate is 50%.

Models	W_1	W_2	nRMSE	Optimality gap (%)
AR model	–	–	34.76%	15.04%
Proposed model	1	20	34.89%	13.91%
	1	10	35.14%	13.42%
	1	5	36.28%	12.71%
	1	2	41.09%	11.88%
	1	1	44.45%	11.44%
	2	1	46.11%	11.38%
	5	1	48.57%	11.38%
	10	1	49.21%	11.36%
	20	1	49.61%	11.36%
	1	0	50.07%	11.36%

when $W_1 = W_2 = 1$ and penalty rate is 50%. The predicted solar power generation of our proposed model is less than AR because the proposed model tries to avoid shortages in order to reduce penalty costs. Therefore, even though our proposed model is less accurate than AR at predicting renewable energy generation, it results in better profit because of the lower penalty cost. As can be seen, the profit related to the AR is less than zero for some hours because the total penalty cost is greater than the total revenue of selling solar energy in those hours.

The proposed method is applied to a penalty cost rate range of 0% to 100% to determine the effect of penalty cost rate on forecasting and optimization results. Figs. 5(a) and 5(b) compare the nRMSE and optimality gap of AR and the proposed model in various values for penalty cost rate when $W_1 = 1$ and $W_2 = 1$. Since AR only minimizes the forecasting error, changes in penalty cost rate do not change its result. Therefore, nRMSE of AR is constant for all penalty cost rates while its shortage cost and optimality gap increase by increasing the penalty rate. In contrast, the penalty cost rate has a significant impact

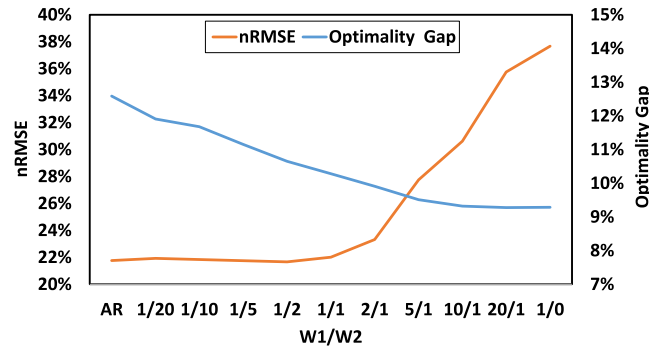


Fig. 6. The nRMSE and optimality gap for MLR and the proposed model when the penalty cost rate is 50%.

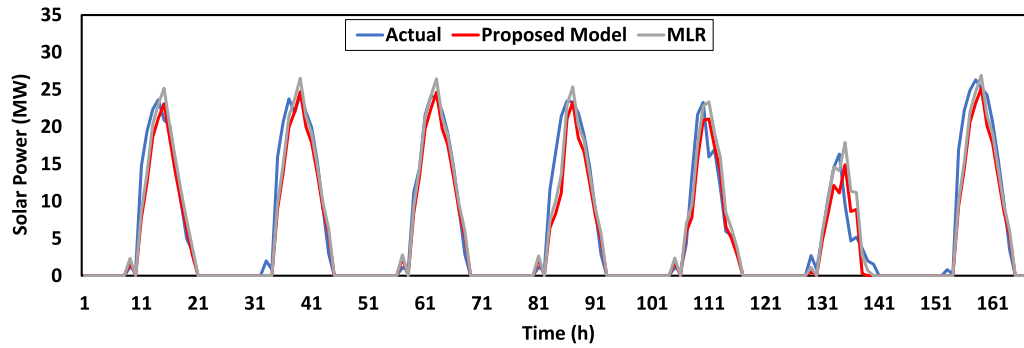
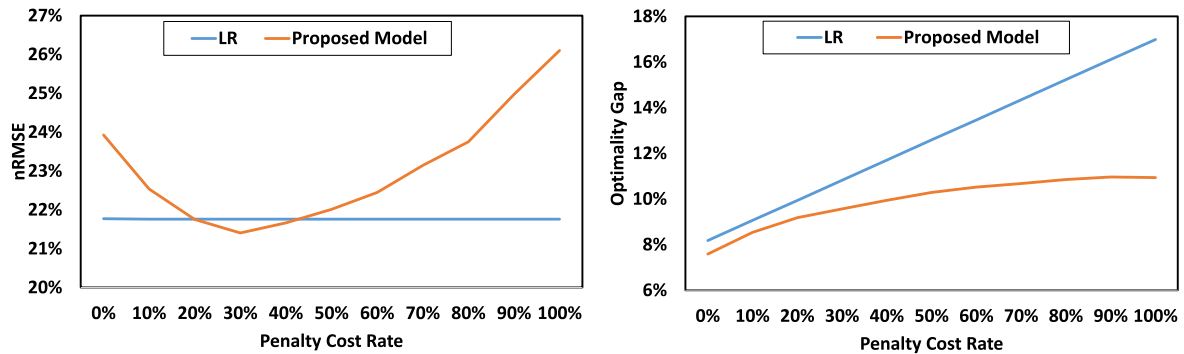


Fig. 7. Actual and predicted solar power generation by MLR and our proposed model when $W_1 = W_2 = 1$ and penalty rate is 50%.



(a) nRMSE of MLR and proposed model in various penalty rate when $W_1 = 1$ and $W_2 = 1$

(b) Optimality gap of MLR and proposed model in various penalty rate when $W_1 = 1$ and $W_2 = 1$

Fig. 8. Comparison of forecasting error and optimality gap of MLR and our proposed model.

Table 4

Comparison between MLR and our proposed model when penalty cost rate is 50%.

Models	W_1	W_2	nRMSE	Optimality gap (%)
Regression model	–	–	21.76%	12.59%
Proposed model	1	20	21.92%	11.91%
	1	10	21.84%	11.68%
	1	5	21.75%	11.15%
	1	2	21.66%	10.65%
	1	1	22.01%	10.28%
	2	1	23.32%	9.91%
	5	1	27.75%	9.51%
	10	1	30.62%	9.32%
	20	1	35.76%	9.27%
	1	0	37.67%	9.28%

on the prediction result of the proposed model. When the penalty rate is 0%, the model increases the commitment much higher than expected solar energy generation because there is no cost for the shortage. This mechanism causes a large forecasting error. For the 10% penalty cost rate, the proposed model reduces the amount of commitment to reduce the risk of shortage because the operator incurs a cost for the shortage. The nRMSE of the proposed model for a 10%–30% penalty rate is close to the AR, which indicates that the value of commitment in this range is close to the expected solar power generation. Penalty rates greater than 30% result in a reduction in commitment. In this case, the commitment is less than expected solar energy generation. Therefore, forecasting error increases by increasing the penalty rate. Because the proposed model changes the energy commitment in response to a change in penalty rate, its results lead to a constant optimality gap. In summary, the proposed model allows renewable energy producers to adjust the amount of solar energy commitment when the penalty cost rate changes, which allows them to maximize profit and minimize

forecasting error at the same time. Without the penalty cost, when the ratio of W_1/W_2 increases, the model significantly increases the commitment to take advantage of the maximum value between the day-ahead and the real-time price, which leads to a high forecasting error.

5.3. Numerical analysis based on exogenous forecasting

For exogenous forecasting, we use a dataset containing solar power generation and day-ahead weather prediction. Multiple MLR and our proposed model are used as exogenous solar power forecasting methods to find a linear relation between day-ahead weather prediction and solar power generation. Parameter selection is one of the most important steps in exogenous forecasting algorithms because using the right parameters will improve the performance of the algorithm and reduce over-fitting. In this paper, we selected 12 independent variables related to the day-ahead prediction of weather information. Also, Month and Hour are considered as other independent variables. We apply a “backward stepwise” algorithm to select the best variables among all 14 variables. According to this method, we first apply MLR to all variables and check the p -value. Then, the variable with the largest p -value or least LogWorth is removed from the list of variables. These steps are repeated until all p -values are less than 0.05. In the last iteration, three variables, SSRD, TSR, and Hour are selected.

All the analyzes in endogenous forecasting are repeated in exogenous forecasting. Table 4 and Fig. 6 shows the results of a comparison between the MLR and our proposed model when the penalty cost rate is 50%. The forecasting error and optimality gap in exogenous forecasting for both MLR and proposed model are better than endogenous forecasting.

Fig. 7 shows actual and predicted values of solar power generation obtained by MLR and our proposed model for 7 sample days when $W_1 = W_2 = 1$ and penalty rate is 50%. The superior performance of exogenous approaches compared to the endogenous approaches in solar power prediction can be observed by comparing Figs. 4 and 7. As with the endogenous method, we gain a better profits when we use our proposed model rather than the MLR because of the reduced penalty cost. Figs. 8(a) and 8(b) indicate the nRMSE and optimality gap of MLR and proposed models for different levels of penalty cost rates. As illustrated, the proposed model's forecasting error decreases until 30% and increases after 30%. Also, in the 30% penalty rate, the proposed model provides better forecasting than MLR. The optimality gap in MLR is linearly proportional to the penalty rate. However, the optimality gap in the proposed model is not proportional to the penalty rate. Like endogenous forecasting, our proposed model adjusts the commitment to maximize profit and reduce the optimality gap in the exogenous approach.

5.4. Discussion on numerical experiment results

Note that numerical experiments were primarily conducted to evaluate the improvement achieved by the proposed approach by comparing it with the traditional decoupled approach where forecasting is decoupled with optimization, and thus, regression models tend to be tuned to solely minimizes forecasting errors without considering the profit driven by the optimization model given forecasted renewable electricity generation. On the other hand, our proposed approach is designed to reflect the feedback from the day-ahead commitment optimized given forecasted renewable electricity generation in the course of regression so that the coefficients of regression model can be fitted to not only minimize forecasting error but also minimize optimality gap. Give this context, we would like to emphasize that the regression model in decoupled approach cannot be easily tuned in a similar way because the impact of forecasting error on day-ahead commitment cannot be directly captured and reflected in regression phase because day-ahead commitment is required to be optimized in response to day-ahead

and real-time price. Although different weights are possibly considered to measure and minimize forecasting errors in different time periods by reflecting the potential loss or penalty cost caused by forecasting errors, those weight vector cannot be easily extracted and applied consistently over uncertain real-time prices. To address this limitation, the proposed approach is designed to directly reflect the feedback from the day-ahead commitment optimized given forecasted solar electricity generation is directly reflected on the course of regression so that both forecasting error and optimality gap can be minimized in a integrated fashion. Because of this aspect, it is possible to tune forecasting model to result in better day-ahead commitment, i.e., increasing profit, despite of the degradation of forecasting accuracy, i.e., increasing forecasting error, as shown in Tables 3 and 4. Moreover, the proposed approach can be tuned with various settings on (i) weights between forecasting errors and optimality gap and (ii) penalty cost rates depending on operators' preference, which cannot be implemented in the traditional decoupled approach. As shown in Figs. 5(b) and 8(b), the optimality gaps tend to be converged at specific level while changing the penalty cost rates. It is reasonably expected that the degree of variation found in renewable electricity generation and the difference between day-ahead and real-time prices would have an impact on the optimality gap. Moreover, profit would be determined by day-ahead commitment driven by forecasted solar electricity generation in response to the deviation between day-ahead and real-time prices, and therefore, the optimality gap would be affected by the deviation found in prices, i.e., the optimality gap would be less if the deviation is less, and it would be higher if the deviation is greater.

6. Conclusion

In this study, we provide a novel technique for predicting day-ahead renewable energy generation in the wholesale energy market's unit commitment problem. This technique integrates the mathematical optimization model with the forecasting model throughout the training phase to minimize the forecasting error and the optimality gap. This research is a pilot study to determine if our proposed approach improves the forecasting result and the optimization model outcome in forecasting-first optimization-second approaches. Regularization parameters are taken into account to maintain a balance between the influence of forecasting error and the optimality gap. The proposed method was applied with endogenous and exogenous forecasting approaches based on AR and MLR models. We performed numerical experiments to evaluate the proposed method's forecasting error and optimality gap compared with traditional forecasting models. The results reveal that, when compared to conventional prediction models, our proposed method reduces the optimality gap by up to 4% for endogenous methods and 3% for exogenous methods. Therefore, it is concluded that the proposed approach results in better forecasting which leads to a higher profit in the optimization problem. Additionally, a sensitivity analysis was performed to determine the effect of the penalty cost rate and regularization parameters on the model. We would like to note that this study would provide preliminary results and insights to apply the proposed approach to various forecasting models including machine learning models. Because the modification of the loss function to reflect the feedback from the optimization model would not be straightforward and might cause computational complexity preventing practical application, the extension of our proposed approach to general forecasting models based on supervised learning will be studied as a future research topic.

CRedit authorship contribution statement

Sajad Karimi: Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Soongeol Kwon:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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